23 S is a geometric sequence.

(a) Given that $(\sqrt{x} - 1)$, 1 and $(\sqrt{x} + 1)$ are the first three terms of S, find the value of x.
You must show all your working.

Equal consecutive ratios:

$$\frac{1}{\sqrt{x} - 1} = \frac{\sqrt{x} + 1}{1}$$

$1 = (\sqrt{x} - 1)(\sqrt{x} + 1)$

1 mark
Equate the ratios and multiply through

1 mark
Express the common ratio

×	$\sqrt{x}$	+1
$\sqrt{x}$	x	$\sqrt{x}$
−1	$-\sqrt{x}$	−1

$1 = x - 1$, so $x = 2$.

1 mark
The middle terms cancel; complete the reasoning

2

(3)

(b) Show that the 5th term of S is $7 + 5\sqrt{2}$

Common ratio $r = \sqrt{2} + 1$.

1 mark
Link the ratio to the fifth term

Fifth term = third term × $r^2 = (\sqrt{2} + 1)^3$.

×	$\sqrt{2}$	+1
$\sqrt{2}$	2	$\sqrt{2}$
+1	$\sqrt{2}$	+1

×	$\sqrt{2}$	+1
3	$3\sqrt{2}$	3
$2\sqrt{2}$	4	$2\sqrt{2}$

First: $(\sqrt{2} + 1)^2 = 3 + 2\sqrt{2}$.

Then: $(3 + 2\sqrt{2})(\sqrt{2} + 1) = 7 + 5\sqrt{2}$.

1 mark
Expand and collect terms to show the fifth term

(2)

(Total for Question 23 is 5 marks)